GROUP PROJECT PROPOSAL

Your Group Number: 6 **Section:** 335.01.02 / 635.02

Your Group's Team Members: Ryan Miner, Myra Mulongoti, & Obumneme Umeonwuka

Project Title: Complaint Resolution Time Prediction

Dataset Name (If decided): 311 Service Requests from 2020 to Present
Link: https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2020-to-Present/erm2-nwe9/about_data

What is the problem statement?

Compare classical machine learning and advanced regression models for predicting resolution time of 311 complaints.

Proposed Methodology (Discuss types of models, validation, evaluation plans):

Models to compare:

Mean baseline
Linear regression
Ridge and lasso regression
Random forest
Gradient boosting (e.g. XGboost)

Validation:

Split the dataset into training, validation, and test sets or use k-fold cross-validation on the training set.
Perform hyperparameter tuning using cross-validation.
Use the same preprocessing and feature set across models to ensure fair comparison.

Evaluation Plans:

We will evaluate the different models' performance on holdout data using the metrics below. We will compare all models against the mean baseline and select the best-performing approach based on the metrics:

Mean Absolute Error (MAE)
Root Mean Squared (RMSE)

GitHub Link (Make sure the repository is public):

https://github.com/rjm5441/MachineLearningFinalProject